INCH-POUND

MIL-DTL-52525/17C
20 November 2008
SUPERSEDING
MIL-DTL-52525/17B
30 June 1998

DETAIL SPECIFICATION SHEET

FITTINGS, CLAMP-HALVES, FLANGE TO PORT, FOR 4-BOLT SPLIT-FLANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525.

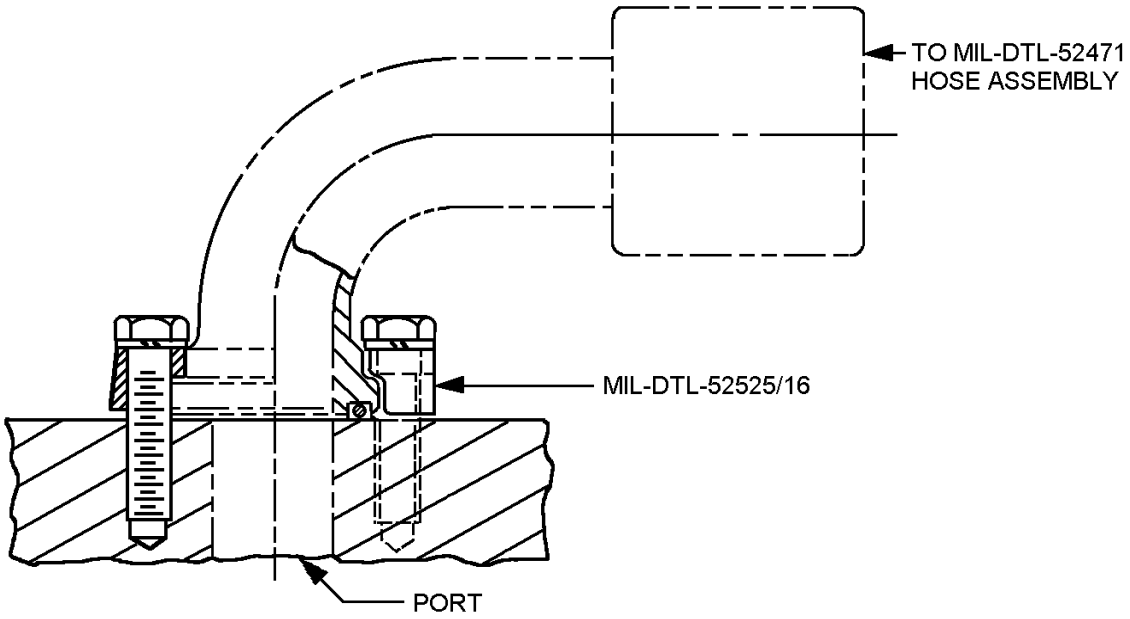
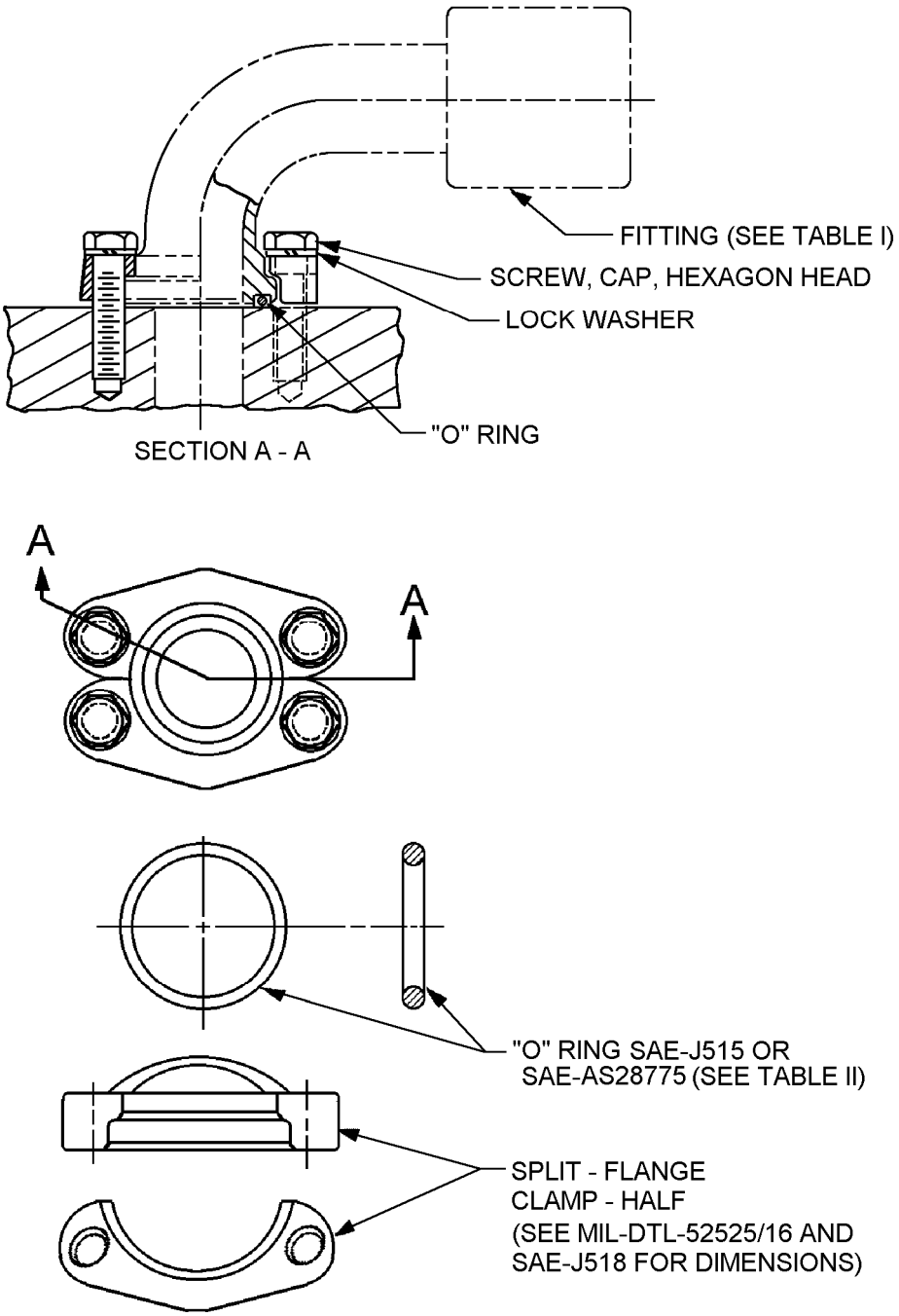


FIGURE 1. Connections: flange to port, 4-bolt split-flange type.

AMSC N/A

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NOTE:
1. The fitting clamp-half flange to port is MIL-DTL-52525/16. This connection is used with MIL-DTL-52471 hose assemblies.

FIGURE 1. Connections: flange to port, 4-bolt split-flange type - Continued.

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TABLE I. Connecting units used with flange type.

Part or Identifying Number (PIN)	Fitting	Screw cap hex head (4 required) NASM90725 PIN 1/ 2/	Washer, lock helical, split (4 required) NASM35338 PIN 1/ 2/	Clamp-halves flange to port	Clamp-halves flange to port SAE-J518 PIN
M52525/17-8	M52525/10-8-8 M52525/11-8-8 M52525/12-8-8	NASM90725 -36	MS35338-45	M52525/16-8	J518-1 8 1161
M52525/17-12	M52525/10-12-12 M52525/11-12-12 M52525/12-12-12 M52525/13-12-12 M52525/14-12-12 M52525/15-12-12	NASM90725 -62	MS35338-46	M52525/16-12	J518-1 12 1161
M52525/17-16	M52525/10-16-16 M52525/11-16-16 M52525/13-16-16 M52525/14-16-16 M52525/15-16-16 M52525/43-16-16 M52525/44-16-16 M52525/45-16-16	NASM90725 -62	MS35338-46	M52525/16-16	J518-1 16 1161
M52525/17-20	M52525/10-16-20 M52525/10-20-20 M52525/11-16-20 M52525/11-20-20 M52525/12-16-20 M52525/12-20-20 M52525/13-20-20 M52525/14-20-20 M52525/15-20-20 M52525/43-20-20 M52525/44-20-20 M52525/45-20-20	NASM90725 -89	MS35338-47	M52525/16-20	J518-1 20 1161
M52525/17-24	M52525/10-20-24 M52525/10-24-24 M52525/11-24-24 M52525/12-20-24 M52525/12-24-24 M52525/13-24-24 M52525/14-24-24 M52525/15-24-24 M52525/43-24-24 M52525/44-24-24	NASM90725 -113	MS35338-48	M52525/16-24	J518-1 24 1161

See footnotes at end of table

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TABLE I. Connecting units used with flange type - Continued.

PIN	Fitting	Screw cap hex head (4 required) NASM90725 PIN 1/ 2/	Washer, lock helical, split (4 required) NASM35338 PIN 1/ 2/	Clamp-halves flange to port	Clamp-halves flange to port SAE-J518 PIN
M52525/17-32	M52525/10-24-32	NASM90725 -113	MS35338-48	M52525/16-32	J518-1 32 1161
	M52525/10-32-32				
	M52525/11-32-32				
	M52525/12-32-32				
	M52525/43-32-32				
	M52525/44-32-32				
	M52525/45-24-24				
	M52525/45-32-32				

1/ Bolts and washers identified in table I are for low carbon steel materials only. If another material listed in table III is used to manufacture bolts or washers the dimensional requirements shall be in accordance with NASM90725 and NASM35338.

2/ Never reuse fasteners after they have been subjected to loading or the elements.

The O-ring legacy tables II and III shows O-rings originally called out by MIL-DTL-52525/17 and shows the superseding PIN for a side by side comparison.

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TABLE II. O-ring legacy replacements for AN6227. 1/ 2/ 3/

Dash size	PIN	Width (T or d2) Inches (mm)	ID (d1) Inches (mm)	Comments
-8	AN6227-15	.0934 (2.37)	.750 (19.05)	Originally MIL-DTL-52525/17 called out the AN6227 and AN6230 PIN's. AN6227 and AN6230 are cancelled and superseded by SAE-AS28775.
-12	AN6227-19	.0934 (2.37)	1.000 (25.40)	
-16	AN6227-24	.125 (3.18)	1.313 (33.35)	
-20	AN6227-27	.125 (3.18)	1.500 (38.10)	
-24	AN6230-3	.125 (3.18)	1.875 (47.63)	
-32	AN6230-6	.125 (3.18)	2.250 (57.15)	
Replacement PIN for the AN6227				
Dash size	PIN	Width (T or d2) Inches (mm)	ID (d1) Inches (mm)	Comments
-8	SAE-AS28775-210	.135-.143 (3.43 - 3.63)	.728 - .0740 (18.49 - 1.880)	The SAE-AS28775 PIN is a replacement for AN6227 and AN6230.
-12	SAE-AS28775-214	.135-.143 (3.43 - 3.63)	.978 - .0990 (24.84 - 2.514)	
-16	SAE-AS28775-219	.135-.143 (3.43 - 3.63)	1.290 - 1.302 (32.76 - 33.070)	
-20	SAE-AS28775-222	.135-.143 (3.43 - 3.63)	1.478 - 1.490 37.54 - 37.846)	
-24	SAE-AS28775-328	.205-.215 (5.21 - 5.46)	1.840 - 1.860 (46.74 - 47.24)	
-32	SAE-AS28775-331	.205-.215 (5.21 - 5.46)	2.215 - 2.235 (56.26 - 56.77)	

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ The width and ID columns are dimensions from the PIN in column 2.

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TABLE III. O-ring legacy replacements for SAE-J120. 1/ 2/ 3/

Dash size	PIN	Width (T or d2) Inches (mm)	ID (d1) Inches (mm)	Comments
-8	SAE-J120-210	.139 ±.004 (3.53 ± 0.10)	.734 (18.64)	SAE-J120 is cancelled without replacement. However the SAE-J120 cancellation notice should state “superseded by SAE-J515”.
-12	SAE-J120-214	.139 ±.004 (3.53 ± 0.10)	.984 (24.99)	
-16	SAE-J120-219	.139 ±.004 (3.53 ± 0.10)	1.296 (32.92)	
-20	SAE-J120-222	.139 ±.004 (3.53 ± 0.10)	1.484 (37.69)	
-24	SAE-J120-225	.139 ±.004 (3.53 ± 0.10)	1.859 (47.22)	
-32	SAE-J120-228	.139 ±.004 (3.53 ± 0.10)	2.234 (56.74)	
Replacement PIN for the SAE-J120				
Dash size	PIN	Width (T, or d2) Inches (mm)	ID (d1) Inches (mm)	Comments
-8	SAE-J515-210	.1389 ±.004 (3.528 ± 0.01)	.734 (18.64)	The SAE-J515 PIN is a replacement for SAE-J120
-12	SAE-J515-214	.1389 ±.004 (3.53 ± 0.10)	.98 (24.89)	
-16	SAE-J515-219	.1389 ±.004 (3.53 ± 0.10)	1.296 (32.92)	
-20	SAE-J515-222	.1389 ±.004 (3.53 ± 0.10)	1.484 (37.69)	
-24	SAE-J515-225	.1389 ±.004 (3.53 ± 0.10)	1.859 (47.22)	
-32	SAE-J515-228	.1389 ±.004 (3.53 ± 0.10)	2.234 (56.74)	

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ The width and ID columns are dimensions from the PIN in column 2.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I and IV.

The O- ring shall have a durometer hardness of 90 ±5.

If fittings are to be used on an oxygen hose, see MIL-DTL-52525.

Materials and finishes shall be in accordance with MIL-DTL-52525 and table I. All finishes shall be capable of withstanding 96 hours minimum of salt spray.

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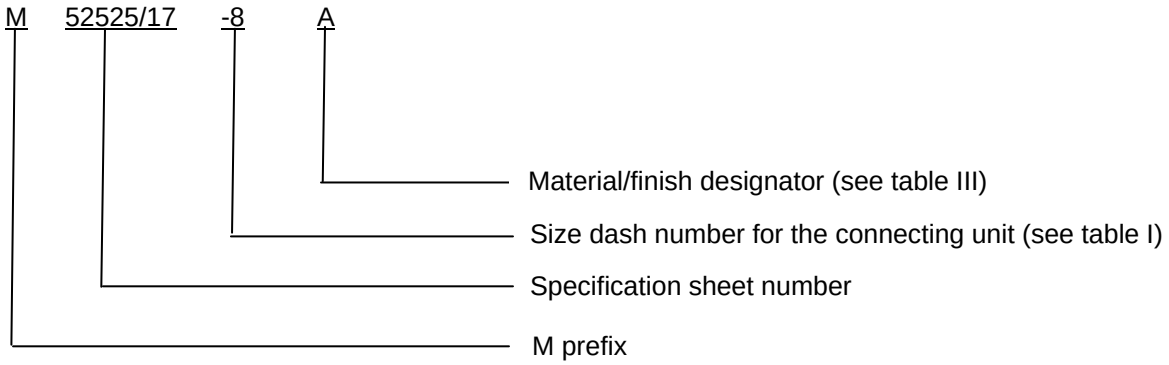
TABLE IV. Material and finish identification codes. 1/ 2/

PIN code material/plating finish	Material	Plating finish
A	Aluminum alloy 6061 or 7075	Anodize in accordance with MIL-A-8625, type II.
AN		Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type 2, class A.
BN		Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
C	Steel	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
CN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YC	Chrome-molybdenum steel alloy 4130	Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2.
YN		Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
F	Steel	NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
FN	Chrome-molybdenum steel alloy 4130	
G	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25.
YG	Chrome-molybdenum steel alloy 4130	
H	Steel	Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. 3/
YH	Chrome-molybdenum steel alloy 4130	
J	Steel	Zinc plating with chromate conversion in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5.
YJ	Chrome-molybdenum steel alloy 4130	
K	Nickel-copper alloy UNS N04400	No additional finish.
M	High-chromium nickel alloy UNS N06690	No additional finish.
N	Steel	Zinc aluminum in accordance with ASTM F1136, grade 3, NC.
YN	Chrome-molybdenum steel alloy 4130	
P	Steel	Zinc plating with colorless passivate in accordance with ASTM B633, type VI, Fe/Zn 5.
YP	Chrome-molybdenum steel alloy 4130	
S	Corrosion resistant steel	No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7.
SN		Passivation above and NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
T	Titanium	Anodize in accordance with SAE-AMS2488, type 2.
V	Steel	Zinc-nickel in accordance with SAE-AMS2417, type 1.
W	Steel	Any zinc plating above.
YW	Chrome-molybdenum steel alloy 4130	
Z	Steel	Any zinc plating above with NAVAIR TCP in accordance with MIL-DTL-81706, type 2, class A.
YZ	Chrome-molybdenum steel alloy 4130	

1/ All materials and finishes shall be in accordance with MIL-DTL-52525.
2/ Embitterment test need not be run.
3/ Zinc phosphate finish is hexavalent chromium free.

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PIN: The PIN consists of the letter “M” the specification sheet number, a dash, a number for the connecting unit, and a material finish designator.



PIN example: M52525/17-8A, describes a .500 inch (12.70 mm) 4-bolt split-flange, aluminum with an anodized finish.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Color identification. Color identification shall be in accordance with SAE-AS4841.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-52525, this document references the following:

MIL-A-8625	ASTM F1136	SAE-AMS2700
MIL-DTL-16232	NASM 35338	SAE-AS4841
MIL-DTL-52471	NASM 90725	SAE-AS28775
MIL-DTL-52525/16	SAE-AMS-C-81562	SAE-J515
MIL-DTL-81706	SAE-AMS-QQ-P-416	SAE-J518
ASTM B633	SAE-AMS2417	
ASTM B695	SAE-AMS2488	

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2007-085)

Review activities:
Army - CR4
Navy - AS, CG, MC, SA, YD

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